

Genome-Wide Associations with MRI Imaging Measures: Hippocampus, Entorhinal, Whole Brain

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Abstract

Dementia due to Alzheimer's Disease (AD) is a common condition which often is reflected in brain volume. Using MRI data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), this study investigates the genetic association between brain volume and brain section's volume in relation to Dementia caused by Alzheimer's Disease. ADNI data, following quality control, included the genetic data of 812 subjects and 1,239,603 autosomal SNPs. A Genome Wide Association Study (GWAS) was used to investigate the linear association between the presence of individual single nucleotide polymorphisms (SNPs) and the MRI imaging measures of whole brain volume, Hippocampus volume, and Entorhinal volume. Our findings are the following: two SNPs, rs769449 and rs4420638, met the significance threshold of $5e-8$ for entorhinal volume and hippocampus volume respectively. In addition to the common covariates: age, sex, and principal components 1-4; we performed a second GWAS for each phenotype with included APOE4 frequency as a covariant. This overall reduced the p-value of each SNP. Some notable SNPs met suggestive thresholds of $1e-5$ in both GWAS including: kgp11369900, kgp2089601, rs12958162, and rs7797507.

Introduction

Alzheimer's disease is the most common form of dementia and is a progressive cognitive disease that affects a person's thought, memory, and language functions. Symptoms of this disease include memory loss, decreased judgment, difficulty completing daily tasks, and changes in personality, mood, or behavior. In 2020, almost 6 million people in the U.S. were diagnosed with Alzheimer's disease, and it is now considered the 6th leading cause of death among the U.S. adult population. However, it has been shown that there is under-reporting of Alzheimer's on death certificates, so the actual proportion of people who die from this disease may be higher than estimated.¹

Because this is such a prevalent disease in the U.S., there has been considerable research done on this topic. Past studies have shown that Alzheimer's disease is associated with entorhinal dysfunction and loss of hippocampal tissue, parts of the brain that are associated with learning, memory, and navigation.² It follows that current studies on Alzheimer's disease have a particular focus on these parts of the brain.

Additionally, the apolipoprotein E4 (APOE4) gene, which is a variant of the APOE gene and aids in the formation of a protein that transfers cholesterol and other fats, is known to increase the risk of AD. 15% to 25% of the population is estimated to have one $\epsilon 4$ allele and 2% to 5% is estimated to have two alleles. Two copies of the $\epsilon 4$ allele are associated with a higher risk of AD than one. However, having the $\epsilon 4$ allele does not necessarily mean an individual will develop AD.³

The outcome of a GWAS is a SNP, which is a genomic variant at a base position in the genome⁴. SNPs are found using a human reference genome.

¹ <https://www.cdc.gov/aging/aginginfo/alzheimers.htm>

² <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9877178/>
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8807768/>

³ <https://www.nia.nih.gov/health/alzheimers-causes-and-risk-factors/alzheimers-disease-genetics-fact-sheet#:~:text=APOE%20CE%B54%20increases%20risk%20for,to%205%25%20carry%20two%20copies.>

⁴ <https://www.genome.gov/genetics-glossary/Single-Nucleotide-Polymorphisms>

Our aim is to find SNPs that are significantly associated with Hippocampus, Entorhinal, and whole brain volume using a GWAS, and what gene they are located on. In addition, we determined whether the APOE4 gene has an effect on Hippocampus, Entorhinal, and whole brain volume when included in the GWAS as a covariate.

Methods

To further investigate the possible effect of genetics on MRI imaging measures, we looked into the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset. There are researchers across 60 different sites in the United States and Canada to collect data on the progression of AD in the human brain in relation to aging, mild cognitive impairment, and dementia.⁵ In order for us to reduce the likelihood of bias in the analysis, the data was filtered so that only variants with minor allele frequencies (MAF) was greater than 5%, missingness for both individuals and variants was less than 5% and our Hardy-Weinberg p-values were greater than 0.000001. From the 2,379,855 total SNPs in the dataset, 1086387 (45%) had < 5% MAF, 20859 (0.876%) had > 5% missingness, and 0.336% had a Hardy-Weinberg p-value of < 0.000001. After removing these SNPs using PLINK on the Shared Computing Cluster, there were 1,239,603 total autosomal SNPs left across the 812 individuals.

Table 1: Distribution of Covariate Classes in Quality Controlled ADNI Dataset

Covariate	Classes	Percentage
Sex	Female	44.8%
	Male	55.2%
Number of APOE4 Alleles	0	59.1%
	1	33.9%
	2	7.02%
Race	American Indian/Alaskan	0.246% (2/812)
	Hawaiian/Pacific Islander	0.246% (2/812)
	White	93.4% (758/812)
	Black	3.45% (28/812)
	Asian	1.35% (11/812)
	More than One	1.11% (9/812)
	Unknown	0.246% (2/812)

A GWAS was conducted using a linear model adjusting for the covariates sex, age, and principal

⁵ <https://adni.loni.usc.edu/about/>

components (PCs) 1-4 generated using Plink. This was done to ensure that the GWAS accurately accounted for population stratification of race by utilizing genetic data. We decided to do a GWAS study so we can identify genes that may be associated with AD. Total brain volume (listed as ICV_bl in the dataset) also was adjusted for when conducting GWAs for the entorhinal and hippocampus phenotypes. Additionally, separate GWAs were done including the APOE4 allele as a covariate for each MRI measure. Since APOE4 has been shown to be one of the strongest genetic risk factors for developing Alzheimer's disease, specifically in those of European ancestry, it must be accounted for in our association tests: people who carry two copies of the APOE4 allele are estimated to have about a 60% chance of developing Alzheimer's disease by the age of 85 and dementia symptoms and diagnosis 7-10 years earlier than those not carrying any APOE4 alleles.⁶

PC1 vs. PC2 of ADNI

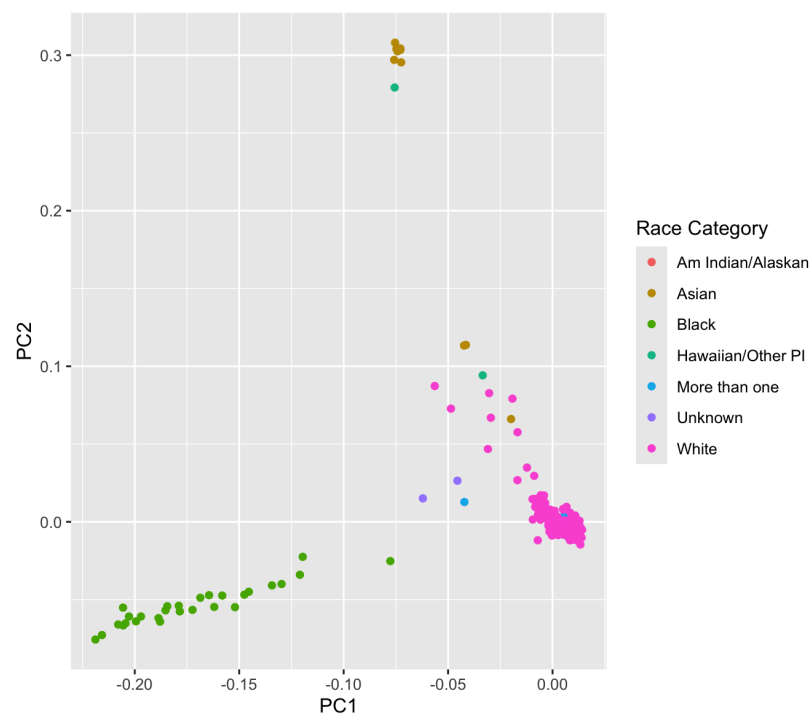


Figure 1: Visualization of PC1 vs PC2

PC3 vs. PC4 of ADNI

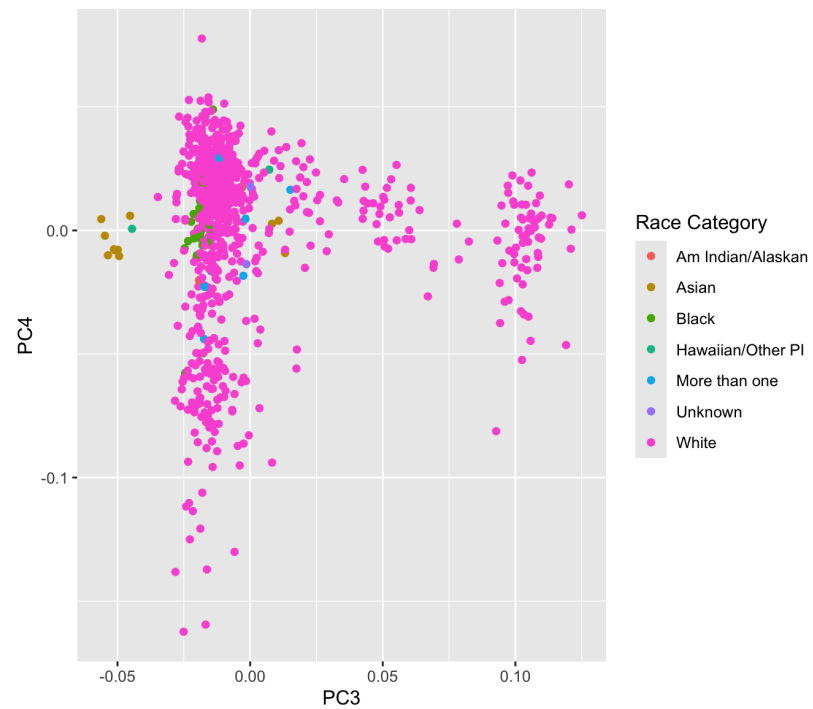


Figure 2: Visualization of PC3 vs PC4

PC5 vs. PC6 of ADNI

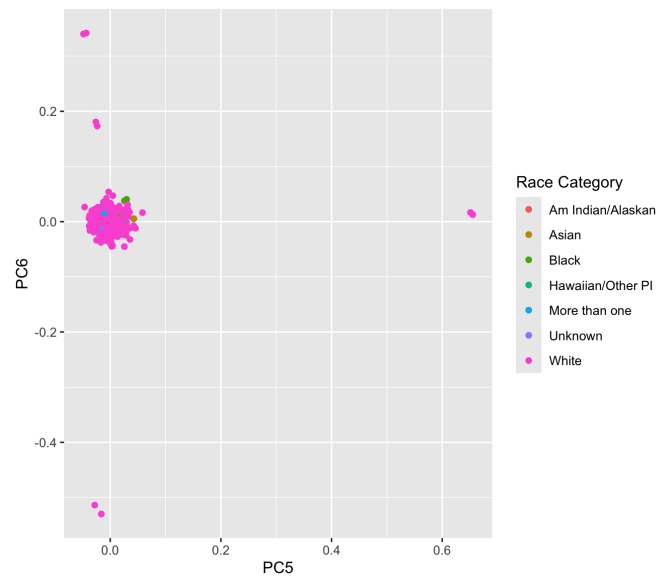


Figure 3: Visualization of PC5 vs PC6

It can be seen from the explicit stratification of the different race categories in Figure 1 that the main principal components account for race. PC3 and PC4 were also used as covariates in the GWA, as there is still slight stratification seen in the population in Figure 2. There is only one main population seen in Figure 3, so PC5 and PC6 were discarded. After importing the PLINK output into RStudio, the qqman R package was used to construct Manhattan and Q-Q plots for each GWAS.

The threshold for a SNP to be considered “suggestive” was defined as a p-value of $1e-5$, and the significance threshold was defined as a p-value of $5e-8$. Additionally, when encountering significant or suggestive SNPs within 500,000 base pairs of each other, the least significant SNP between the two was excluded in order to reduce potential linkage disequilibrium that may be present.

Results

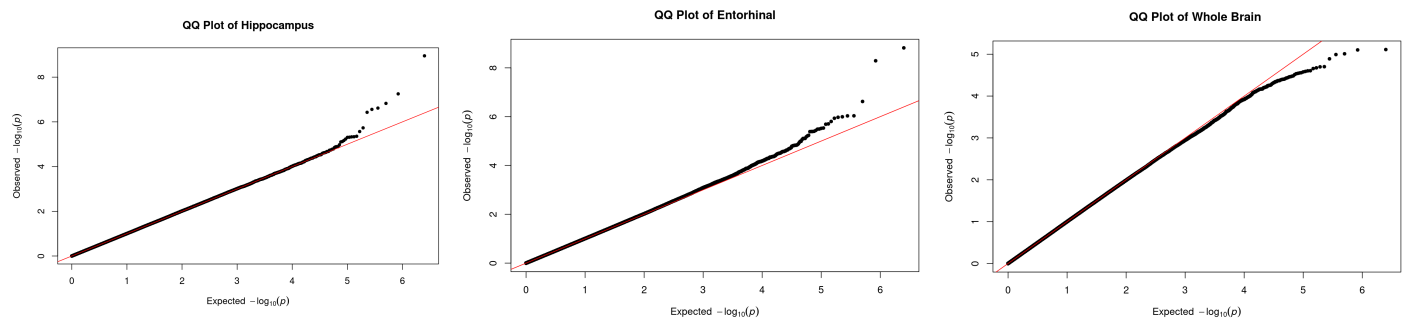


Figure 4: Q-Q Plots for Hippocampus, Entorhinal, and Whole Brain MRI Imaging Measures (adj. sex, age, PCs 1-4)

The QQ plots generated for each phenotype look typical, with the majority of the observed p-values being aligned with the expected p-values and the smaller p-values deviating from what was expected.

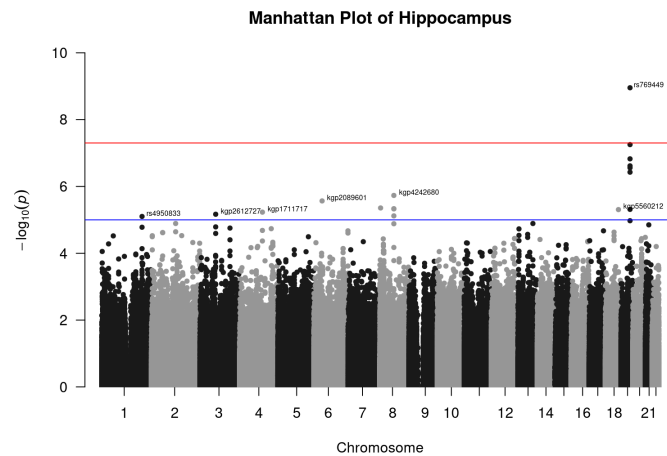


Figure 5: Manhattan Plot of Hippocampus (adj. sex, age, total brain volume, PCs 1-4)

Adjusting for age, sex, total brain volume, and PC1, PC2, PC3, and PC4; SNP rs769449 on chromosome 19 with effect allele A showed to be significantly associated with volume of the Hippocampus with a p-value of $1.11\text{e-}09$. For each copy of this allele, there is a predicted 399.100 unit decrease in Hippocampus volume.

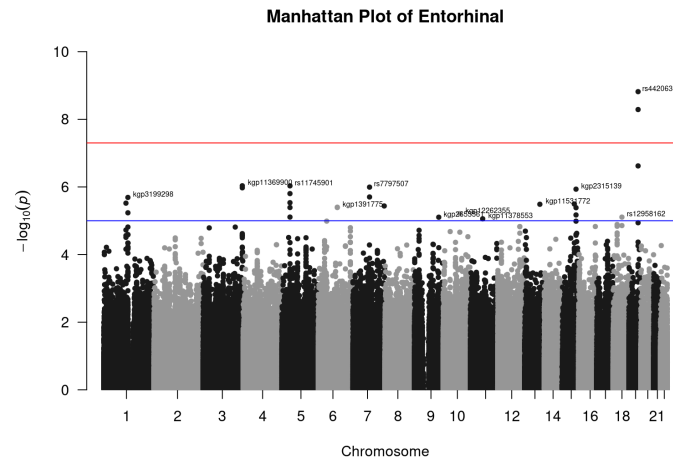


Figure 6: Manhattan Plot of Entorhinal (adj. sex, age, total brain volume, PCs 1-4)

Additionally, after adjusting for the same covariates, SNP rs4420638 on chromosome 19 with effect allele G was observed to be associated with Entorhinal volume with a p-value of $1.52\text{e-}9$. For every copy of this allele, Entorhinal volume is predicted to decrease by 239.5 units.

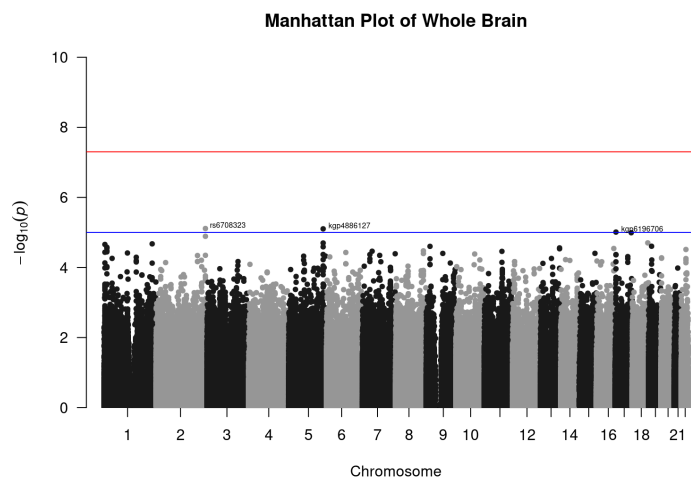


Figure 7: Manhattan Plot of Whole Brain (adj. sex, age, PCs 1-4)

After adjusting for covariates, there are no genome-wide significant SNPs for whole brain.

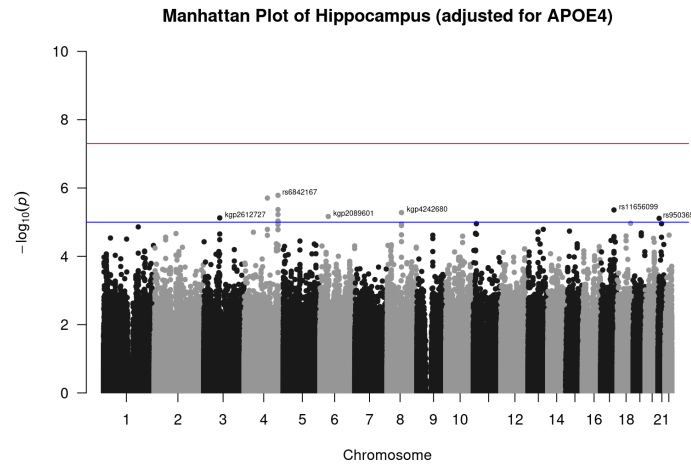


Figure 8: Manhattan Plot of Entorhinal (adj. sex, age, total brain volume, PCs 1-4)

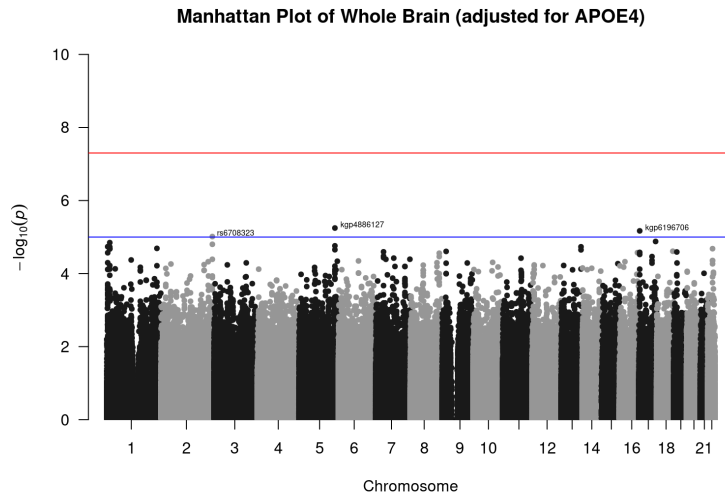
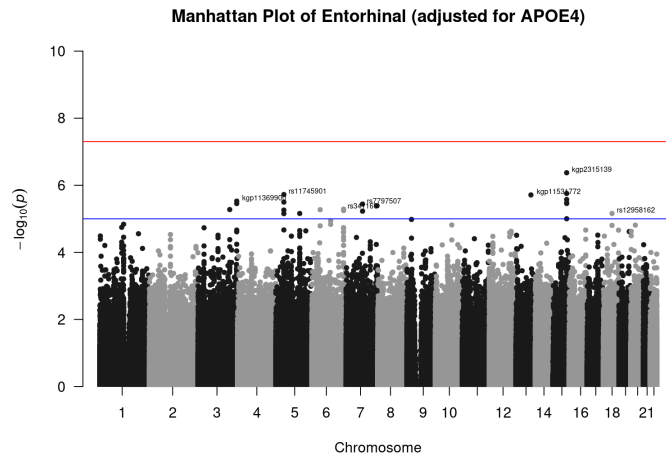


Figure 9: Manhattan Plots of Hippocampus, Entorhinal, Whole Brain (adj. sex, age, PCs 1-4, APOE4)

After adjusting for age, sex, PC1, PC2, PC3, PC4, and the APOE4 allele, no SNPs were found to be significantly associated with any of the MRI phenotypes.

Table 2: Statistically Significant SNPs

CHR	SNP	BP	A1	NMISS	BETA	STAT	P	Phenotype
19	rs769449	45410002	A	720	-399.1	-6.174	1.12E-09	Hippocampus
19	rs4420638	45422946	G	713	-239.5	-6.123	1.52E-09	Entorhinal

Table 3: Suggestive SNPs

CHR	SNP	BP	A1	NMISS	BETA	STAT	P	Phenotype
1	rs4950833	202159922	G	715	275	4.502	7.88E-06	Hippocampus
3	kgp2612727	78360853	C	720	366.3	4.534	6.79E-06	Hippocampus
4	kgp1711717	114248646	C	720	-522.5	-4.566	5.86E-06	Hippocampus
6	kgp2089601	41310031	C	719	548.3	4.73	2.71E-06	Hippocampus
8	rs1647315	4066831	C	720	-270.0	-4.627	4.403E-06	Hippocampus
8	kgp4242680	70409280	C	719	253	4.808	1.86E-06	Hippocampus
18	kgp5560212	65218751	A	717	-411.5	-4.602	4.95E-06	Hippocampus
3	kgp2612727	78360853	C	720	353.3	4.512	7.50E-06	Hippocampus (APOE4)
4	kgp1711717	114248646	C	720	-531.0	-4.796	1.969E-06	Hippocampus (APOE4)
4	rs6842167	166370111	T	720	296.2	4.834	1.64E-06	Hippocampus (APOE4)
6	kgp2089601	41310031	C	719	510.4	4.534	6.80E-06	Hippocampus (APOE4)
8	kgp4242680	70409280	C	719	234.7	4.59	5.25E-06	Hippocampus (APOE4)
17	rs11656099	64793662	A	720	-257.6	-4.628	4.40E-06	Hippocampus (APOE4)
21	rs950365	20212962	T	720	-231.3	-4.506	7.74E-06	Hippocampus (APOE4)
1	kgp3199298	120023971	C	692	-199.1	-4.79	2.04E-06	Entorhinal
3	kgp11369900	193586745	A	718	-181	-4.95	9.28E-07	Entorhinal
5	rs11745901	40627066	C	718	180.5	4.95	9.27E-07	Entorhinal
6	kgp1391775	94064757	T	718	-333	-4.648	3.99E-06	Entorhinal
7	kgp6173352	155282007	T	713	171.7	4.667	3.653E-06	Entorhinal
7	rs7797507	82698998	G	718	361.4	4.933	1.01E-06	Entorhinal
9	kgp2655561	119742974	C	718	-189.8	-4.503	7.85E-06	Entorhinal
10	kgp12262355	83019732	G	717	-239.7	-4.555	6.17E-06	Entorhinal
11	kgp11378553	60723422	G	718	171.7	4.478	8.78E-06	Entorhinal
13	kgp11531772	94122100	A	718	361.6	4.691	3.26E-06	Entorhinal
15	kgp2315139	89688115	T	718	174.2	4.904	1.17E-06	Entorhinal

18	rs12958162	43249693	A	718	-168.7	-4.505	7.77E-06	Entorhinal
2	rs992089	158684240	T	718	166.0	4.588	5.297E-06	Entorhinal (APOE4)
3	kgp11369900	193586745	A	718	-168.4	-4.71	2.99E-06	Entorhinal (APOE4)
5	rs11745901	40627066	C	718	171	4.806	1.88E-06	Entorhinal (APOE4)
5	kgp12224359	119932968	T	718	182.2	4.530	6.921E-06	Entorhinal (APOE4)
6	kgp434743	41639575	A	717	163.4	4.586	5.344E-06	Entorhinal (APOE4)
6	rs341160	158832854	G	718	-162	-4.594	5.15E-06	Entorhinal (APOE4)
7	rs7797507	82698998	G	718	334.6	4.67	3.61E-06	Entorhinal (APOE4)
7	kgp6173352	155282007	T	713	166.7	4.643	4.09E-06	Entorhinal (APOE4)
13	kgp11531772	94122100	A	718	360.2	4.799	1.94E-06	Entorhinal (APOE4)
15	kgp2315139	89688115	T	718	176.5	5.106	4.24E-07	Entorhinal (APOE4)
18	rs12958162	43249693	A	718	-165.3	-4.531	6.88E-06	Entorhinal (APOE4)
2	rs6708323	235110179	T	791	-23290	-4.502	7.74E-06	WholeBrain
5	kgp4886127	164867020	T	796	-20790	-4.497	7.92E-06	WholeBrain
17	kgp6196706	3454669	A	790	23420	4.451	9.77E-06	WholeBrain
2	rs6708323	235110179	T	791	-22970	-4.453	9.71E-06	WholeBrain (APOE4)
5	kgp4886127	164867020	T	796	-21050	-4.57	5.67E-06	WholeBrain (APOE4)
17	kgp6196706	3454669	A	790	23760	4.532	6.76E-06	WholeBrain (APOE4)

Discussion

In our results, rs769449 was found to be significantly associated with the Hippocampus, and Hippocampal atrophy is associated with AD. In Yashin et. al, Alzheimer's disease was found to be associated with the rs769449 SNP. When adjusting for APOE4, there were no significant associations. For the entorhinal cortex, rs4420638 was found to be significantly associated with this phenotype, and Entorhinal cortex atrophy is associated with AD. However, when adjusting for APOE4 as a covariate, there were no significant associations. Since rs769449 is located on the APOE gene, rs4420638 is located on the APOC1 gene which is in the APOE gene cluster, and both are on chromosome 19, this suggests that APOE has an effect on both Hippocampus and Entorhinal volumes and explains why there were no significant associations when APOE4 was included as a covariate. APOE is the gene that is associated with the Hippocampus and Entorhinal cortex, so without accounting for APOE4, the SNPs seem to be associated with their respective phenotypes when it is APOE that is influencing this association. In four studies, it was concluded that APOE4 carriers with AD have greater Hippocampal and Entorhinal cortex atrophy than individuals with AD and no APOE4 alleles, which supports our results.

Suggestive SNPs should be looked into further in future studies. Additionally, more efforts should be made to increase the diversity of future Alzheimer's Disease datasets and recruit from non-white populations in order to have a better understanding of genetic risk factors in those populations. Below, it can be seen that the ADNI dataset is dominated by white participants and each non-white group has less than a seventh of the number of white participants.

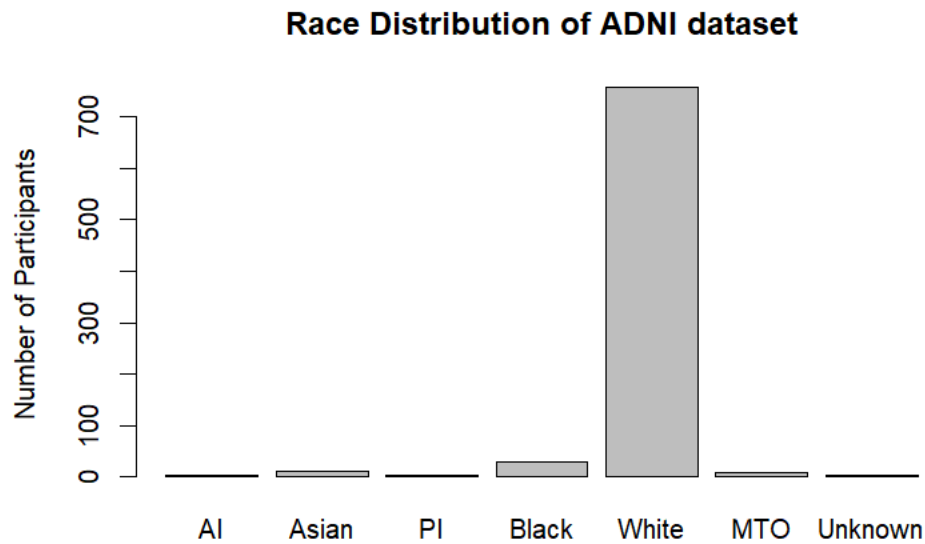


Figure: Bar plot of race distribution of the quality controlled ADNI dataset
Abbreviations: AI: American Indian/Alaskan, PI: Hawaiian/Pacific Islander, MTO: More than One

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